

Arjun Rahar

arjunrahar@myjobemails.com | +1 (404) 203-9386 | Georgia, USA | [LinkedIn](#) | [GitHub](#)

Summary

Software Engineer with 4+ years of experience building high-performance systems across robotics, AI, and full-stack engineering. Skilled in Python, C++, TypeScript, microservices, ML pipelines, reinforcement-learning control, ROS-based autonomy, and real-time perception. Developed scalable APIs, model-driven automation, multi-robot coordination engines, and simulation frameworks accelerating validation cycles. Adept at integrating cloud architectures, telemetry pipelines, and production-grade deployment workflows to deliver resilient, data-driven, and intelligence-powered software solutions across complex, safety-critical environments.

Skills

Programming & Scripting: Python, C++, TypeScript, JavaScript, Go, SQL, secure coding patterns, asynchronous workflows, modular automation, API integration, parallel computing, scripting optimization, Git, GitHub, GitLab, CI/CD workflows

AI & Machine Learning Engineering: PyTorch, TensorFlow, scikit-learn, transformer architectures, diffusion models, LLM fine-tuning, retrieval design, vector databases, reinforcement-learning control, OpenAI Gym simulations, data augmentation, MLOps pipelines, model optimization, structured evaluation

Backend & Systems Engineering: FastAPI, Flask, Django, Node.js, Express.js, microservices, event-driven orchestration, PostgreSQL, MongoDB, Redis, authentication systems, scalable compute design, API gateway management, caching strategies, distributed processing

Frontend & UI Engineering: React, Next.js, TypeScript-driven state patterns, component libraries, responsive UI engineering, WebGL interfaces, accessibility compliance, ML-integrated UI flows, performance profiling, testing frameworks

Robotics & Autonomous Systems: ROS1, ROS2, LiDAR mapping, EKF localization, SLAM pipelines, multi-robot coordination, motion-planning algorithms, trajectory optimization, actuator control, sensor fusion, real-time constraints, robotic telemetry handling

Cloud, DevOps & Infrastructure: AWS, Azure, Kubernetes, Docker, Terraform, load balancing, observability tooling, distributed logging, secrets management, system reliability workflows, serverless execution, deployment automation, cost optimization

Professional Experience

Founding Software Engineer, *Mira Mace*

May 2025 – Present | Remote, USA

- Owned the full-stack development of the marketing-to-product pipeline, designing responsive frontend landing pages for ad campaigns and backend API endpoints to capture, validate, and persist leads into Supabase.
- Architected and implemented a custom internal CRM and analytics platform using Retool, integrating Supabase queries to deliver real-time dashboards for monitoring marketing, patient data, and lead management processes.
- Built Python backend services leveraging Gemini and OpenAI models to automate patient profile creation and personalized care action plan generation, orchestrating workflows and persisting data efficiently in Supabase.
- Engineered backend integration for a Livekit voice agent deployed on Azure, developing Python APIs to seamlessly connect the voice communication system with the core application and workflow pipelines.

AI Engineer, *Magna International*

May 2024 – Dec 2024 | Michigan, USA

- Built a Flask-based parts tracking microservice for automotive assembly lines, integrating secure RFID data ingestion, PostgreSQL optimization, and asynchronous processing to reduce component mismatch incidents across plants.
- Created a React dashboard for real-time defect visualization in chassis manufacturing, using WebSockets, sensor filtering, and dynamic components to accelerate inspection decisions by 31% across production teams.
- Built a predictive maintenance model using Python, Scikit-learn, and vibration telemetry, forecasting failure risks for robotic welding systems, reducing unplanned downtime and significantly improving maintenance scheduling accuracy.
- Implemented a YOLO-based computer vision model for paint defect classification on automotive panels, exposing inference through Flask APIs, improving detection and reducing manual inspections in finishing lines.

Robotics Software Engineer, *Larsen & Toubro*

Aug 2020 – Jul 2023 | Mumbai, India

- Designed autonomous navigation logic for a construction-site rover using ROS1 navigation stack and LiDAR mapping, improving obstacle negotiation accuracy by 34% and enabling stable multi-terrain traversal across three active infrastructure zones.
- Integrated real-time motor controllers with EKF-driven localization and ultrasonic depth sensors, enhancing path-tracking precision by 22% while maintaining stable loop-timing constraints under heavy dust-load conditions typical in L&T deployment sites.
- Built a reinforcement-learning-based excavation-arm control module using OpenAI Gym simulations, achieving 18% smoother bucket articulation and reducing hydraulic strain cycles by optimizing trajectory profiles for varying soil-density conditions.
- Developed a multi-robot task-allocation engine enabling synchronized fleet operations for material transport, improving inter-robot coordination efficiency by 27% and automating previously manual routing workflows within controlled construction environments.
- Engineered Python microservices handling telemetry normalization, anomaly detection pipelines, and configuration syncing across robotic units, ensuring consistent field-runtime states through data-parsing and event-queue mechanisms.
- Built Python tooling for automated simulation scenario generation, accelerating validation cycles for new motion-planning algorithms by 41% and enabling reproducible test matrices across multiple simulated construction-site conditions.
- Implemented Python-driven deployment scripts and structured logging modules for robot firmware updates, standardizing upgrade cycles across three hardware families and improving debugging turnaround through refined diagnostic-event traceability.

Education

Master of Science in Computer Science, Georgia Institute of Technology

Aug 2023 – May 2025 | Georgia, USA

Bachelor of Technology in Aerospace Engineering, Indian Institute of Technology

Jul 2016 – Jul 2021 | Mumbai, India